

Unit 1, Lesson 10: Multiplying and Dividing Radical Expressions



Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.



Learning Target

- ☐ I can multiply and divide two square roots or two cube roots.



1.10.1 Warm-Up: True Equations

- Using the digits 0 to 9, place a digit in each blue box to make both equations true. Each digit can only be used one time, at most.

$$\sqrt{\square\square} = \square\sqrt{\square}$$

$$\sqrt{\square\square} = \square$$



1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Recall the **quotient of powers** law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

Ariana used the **quotient of powers** law of exponents to simplify $\frac{\sqrt{10}}{\sqrt{10}}$. Her work is shown.

$$\frac{\sqrt{10}}{\sqrt{10}} = \frac{10^{\frac{1}{2}}}{10^{\frac{1}{2}}} = 10^{\frac{1}{2} - \frac{1}{2}} = 10^0 = 1$$

When dividing radicals with the same index, the **quotient law of radicals** is used to rewrite the expression. Each of the following equivalent expressions demonstrates the quotient law of radicals.

$$\frac{\sqrt{13}}{\sqrt{13}} = \sqrt{\frac{13}{13}} = \sqrt{1} = 1 \qquad \frac{\sqrt{21}}{\sqrt{21}} = \sqrt{\frac{21}{21}} = \sqrt{1} = 1$$

1. Rewrite $\frac{\sqrt[3]{2}}{\sqrt[3]{2}}$ using the quotient law of radicals.

When dividing radical expressions, rewrite radical expressions in equivalent forms and use the quotient law to eliminate radicals where possible.

2. Write an equivalent expression for $\frac{\sqrt{125}}{\sqrt{245}}$.

$\frac{\sqrt{125}}{\sqrt{245}}$	
$\frac{\sqrt{125}}{\sqrt{245}} = \frac{\sqrt{\quad \cdot 5}}{\sqrt{\quad \cdot 5}}$	The radicands _____ and _____ are rewritten as the product of a factor and 5.
$\frac{\sqrt{\quad \cdot 5}}{\sqrt{\quad \cdot 5}} = \frac{\quad \cdot \sqrt{5}}{\quad \cdot \sqrt{5}}$	The radicals are rewritten using the _____ law of radicals.
$\frac{\quad \cdot \sqrt{5}}{\quad \cdot \sqrt{5}} = \frac{\quad}{\quad} \cdot \frac{\sqrt{5}}{\sqrt{5}}$	Rewrite each perfect square root.
Because $\frac{\sqrt{5}}{\sqrt{5}} = 1$, $\frac{\quad}{\quad} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\quad}{\quad}$.	

3. With your partner, rewrite the expression $\frac{\sqrt{275}}{\sqrt{44}}$ using the quotient laws of radicals.

4. Jean and Marcus were asked to explain how they would multiply $\sqrt{8} \cdot \sqrt{8}$. Their explanations are shown.

Jean's Explanation	I would first rewrite the radical expressions using rational exponents and then use the product of powers law of exponents to multiply the like bases. $\sqrt{8} \cdot \sqrt{8} = 8^{\frac{1}{2}} \cdot 8^{\frac{1}{2}} = 8^{\frac{1}{2} + \frac{1}{2}} = 8^{\frac{2}{2}} = 8^1 = 8$
Marcus's Explanation	I would use the product law of radicals to multiply the radical expressions and then use the law again to rewrite the square root. $\sqrt{8} \cdot \sqrt{8} = \sqrt{8 \cdot 8} = \sqrt{64} = 8$

Do you agree that both expressions have a value of 8?

5. Explain which method you would use to multiply $\sqrt{12} \cdot \sqrt{12}$.
6. Multiply the following radical expressions.

$$\sqrt{6} \cdot \sqrt{18}$$



In cases where there is a fraction with a radical in the **denominator**, mathematicians use a technique called **rationalizing the denominator** to rewrite an expression.

A model of how to rationalize the expression $\frac{30}{\sqrt{2}}$ is shown.

$$\frac{30}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{30\sqrt{2}}{\sqrt{4}} = \frac{30\sqrt{2}}{2} = 15\sqrt{2}$$



1.10.3 Activity: Pass Around

In groups of four, each student completes the **first** step of their problem. The students then pass their problem to the student on their right to complete the next step. For each new step, continue passing the problems to the student on the right.

Student A $\frac{\sqrt[3]{200}}{\sqrt[3]{625}}$	Student B $\sqrt[3]{14} \cdot \sqrt[3]{20}$
Student C $\sqrt{24} \cdot \sqrt{30}$	Student D $\frac{\sqrt{256}}{\sqrt{108}}$



1.10.4 Your Turn!

- Write an equivalent expression for each radical expression.

a. $\sqrt{9} \cdot \sqrt{9}$

b. $\frac{\sqrt[3]{56}}{\sqrt[3]{77}}$

c. $\frac{2\sqrt{22}}{\sqrt{8}}$

- Anjelica was asked to rewrite the expression $\frac{\sqrt[3]{264}}{\sqrt[3]{81}}$. Anjelica's work is shown. Write an explanation to Anjelica describing why her problem is incorrect. Then, show how to correctly work the problem.

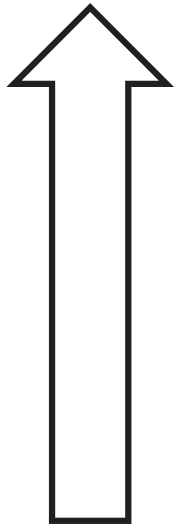
$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{9 \cdot 9}} = \frac{2\sqrt[3]{33}}{9}$$



1.10.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the following.

I can multiply and divide two square roots or two cube roots.



I am so confident; I could explain this to someone else!

I can get to the right answer, but I do not understand well enough to explain it yet.

I understand some of this, but I do not understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will get help by ...

I do not understand any of this yet. I will get help by

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